SHAHIL KADIA

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SUMMARY

AI System Architect (Consultant) at a multi-tenant AI-SEO SaaS on AWS ECS Fargate. Since 2025 I have shipped LLM systems to Gulf government clients and CLEO: RAG (retrieval-augmented generation) and a fail-closed channel of AI-opened pull requests. Recent work secures production agents, metric series down **~94%**, alarm queries **~72x** cheaper.

PROFESSIONAL EXPERIENCE

RegenAI (Remote) **Jan 2025 - Present**

AI products for Gulf government clients and CLEO, a multi-tenant AI-SEO SaaS (FastAPI, React/TypeScript, multi-region ECS Fargate)

AI System Architect (Consultant) **Aug 2026 - Present**

- Eliminated 5 verified remote code execution (RCE) chains in AI-generated WordPress code via a context-typed stdlib-only escaper at each sink, backed by 541 payloads + fuzzing against a ported PHP parse oracle. In production.
- Made LLM arguments untrusted. Python stack-frame analysis strips auto-apply flags, fails closed on unwalkable stacks. Identity is server-pinned, writes need a human, each run budgeted. 41,043 tests, zero regressions.
- Centralized prompt-injection defense on grammar invariants: regex value-pinning that cut a 62.9s ReDoS to 0.009s, unicode-safe fence neutralization, and a working repro of the composition bypass that closed the marker forgery holes.
- Designed disaster recovery (DR) with Multi-AZ, cross-region replica, backups, split-brain fencing on forty-five Terraform resources. Soak: 168/168 hours, zero breaches, 92s lag, recovery time objective (RTO) 10-15 min (drill).
- After classifying **462,901** requests (~1.4% flagged), moved production AWS web application firewall (WAF) from count to block, byte-exact rollback pre-built. ~470 blocked beacons traced to count-mode logging that kept only the first rule.
- Cut CloudWatch series **~94%** (5,679 to ~330) by moving task ids from EMF (embedded metric format) dimensions to properties, so deploys cannot spawn series. Alarm queries retired at **~72x** lower cost, no coverage loss.

AI Developer Intern **Jan 2025 - Jul 2026**

- Built the RAG knowledge stack for a Gulf government client: pgvector hybrid search, semantic chunking, per-tenant sharded vector tables, and a cross-lingual Arabic-English re-query strategy for bilingual document QA.
- Found the cause of a recurring WebSocket outage: query-string auth JWTs tripped AWS WAF managed rules. Moved authentication into RFC 6455 subprotocol bearer tokens with a backend-first rollout, unblocking WAF block mode.
- Created a GitHub App verification ladder for AI-opened pull requests (draft, verify, then promote or discard, infra vs code faults classified), so a PR reaching production has already built green, proven on a public demo site.
- Operated vLLM FP8 MoE serving at 92k-token context (tool/reasoning parsers, fused-MoE kernels), co-serving four models on one 8x L20 host with per-GPU assignment on Alibaba Cloud (MoE FP8, AWQ, Arabic LLM, embeddings).
- Wrote inference benchmarks and topology documentation, measuring TTFT/ITL (time to first token, inter-token latency) P50-P99 with bootstrap intervals and 10,000-request throughput tests. FP8-vs-AWQ calls came from these numbers.
- Routed across 11 model classes (self-hosted GLM via a custom provider) and built a hybrid thinking router whose bilingual complexity classifier decides per request when to think, so paid reasoning only runs on complex queries.

TECHNICAL PROJECTS

voxbar: privacy-first macOS dictation, Rust/Tauri (open-source fork extended) | github.com/nexiouscaliver

- Extended an open-source dictation engine into a product line through v1.6.2, spanning local-LLM post-processing and a LAN TLS phone companion, with 29 verified modules behind release gates.
- Shipped on-device transcript polish: GGUF inference in an isolated worker, grammar-constrained structured outputs (JSON), sha256-pinned models. An exclusive-swap state machine keeps voice model and LLM from sharing RAM.
- Delivered nineteen releases in five days (~4/day peak), each byte-verified by thirteen gates and a 944-function suite. Largest cycle: 9h41m, forty-nine subagents, ~625M tokens, unattended. Used daily by a small group of early users.

OmniForge: self-hosted MR review and merge automation (Claude Code plugin + MCP) | 19 releases

- Designed a 23-tool Model Context Protocol (MCP) server for the merge-request lifecycle (gather, isolate, review, adjudicate, approve).
- Put three review agents with non-overlapping lenses (process, architecture, security) in isolated worktrees behind a fail-closed merge gate: no unpinned approvals, no unresolved threads, approval bound to the checked head commit.
- In production at RegenAI, holding median code-review turnaround at 26 minutes across a 7-wave campaign.

mission-control: multi-session agent-program orchestration | 14 releases in 3 weeks

- Built the controller that runs multi-week agent programs: waves and lanes, build-parallel/enable-serial ordering, parallelism capped by available resources, with agents doing the implementation.
- Made reality checks the default for agent programs: every claim is checked against machine records and live state, and a blocked agent reports the block instead of improvising. Six agent-runtime defects became standing conventions.
- Validated a nightly knowledge-vault consolidation by holdout evaluation (51 of 51, all 16 negatives rejected) before unattended runs. A global lease guard now caps fleets after OOM incidents (peak RSS ~1.5GB to 580MB).

AmiScheduler: university venue-booking platform (HTML/CSS/JavaScript/Flask/PostgreSQL) | github.com/nexiouscaliver/amity-booking-portal

- Booking platform for 15 venue halls (35+ faculty daily, 200+ bookings a month) with no downtime and automated approvals.
- Recognized by the Vice Chancellor for outstanding contribution to university infrastructure. Deployed on the university intranet as a core operational application.

Open Source: 5 merged third-party PRs - migrated mem0 examples to the v3 filters API, made the Dify db.session explicit, fixed a CAMEL MCPToolkit connection bug, extended the pyWhat IPv6 regex to hex, corrected zimfarm worker-dashboard sizing.

TECHNICAL SKILLS

- **AI/LLM Engineering:** Multi-agent orchestration, Agno (multi-agent framework), multi-LLM API routing with tool calling (OpenAI, Anthropic, GLM, DeepSeek), pgvector (hybrid and semantic search), vLLM GPU inference ops (L20/H20)
- **Languages & Frontend:** Python (2 yrs daily production), TypeScript/JavaScript, Rust (shipped product), SQL, Bash, React / MUI / Redux / Vite, Astro
- **Backend:** FastAPI (146 route modules), production-scale WebSocket reliability (custom gunicorn worker, subprotocol auth), SQLAlchemy 2 / Alembic, PostgreSQL, MongoDB, Redis/Valkey, Stripe, Auth0
- **Cloud & IaC (infrastructure as code):** AWS ECS Fargate (multi-region), event-driven SQS / EventBridge / Lambda, ALB (application load balancing), CloudWatch metrics (EMF) + FinOps, Terraform, GitLab CI/CD, Docker containers
- **Security & Observability:** AppSec (SSRF / XSS / RCE), prompt-injection defense, SOC 2 readiness (SAST, dependency scans), Sentry (self-hosted), LLM observability & tracing (Langfuse), OpenTelemetry / Prometheus
- **AI-Assisted Engineering:** MCP server authorship, Claude Code plugin/skill authorship, pytest testing at 41k scale

EDUCATION & CERTIFICATIONS

B.Tech, Computer Science & Engineering, Amity University Mumbai **2022 - 2026**
CGPA 9.20/10, First Class with Distinction, all 8 semesters, no failed courses

Certification: AWS Academy Graduate, Cloud Developing, Dec 2024

Early recognition: GDSC CyberSecurity Lead (50+ students), Smart India Hackathon state qualifier